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Real-Time Music Visualization with Isochords

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Introduction

- New music visualization techniques are valuable for harmonic analysis
  - In Western music theory, most harmony is represented using a piano keyboard or sheet music
- Isochords [1] are a promising alternative
  - Constructed over a grid made of six staggered vertical rows of parallel fifths [1]
  - Clearly highlight the harmonic structure of a musical piece [1]

Objective

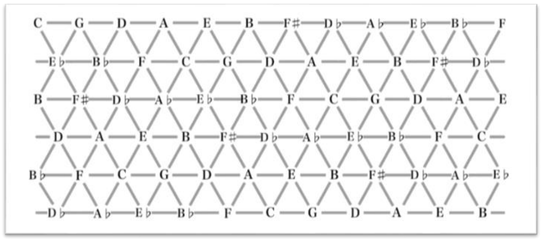
We aim to use an isochord grid, constructed in Unity, to create a real-time visualization of harmonic material contained in a MIDI file.

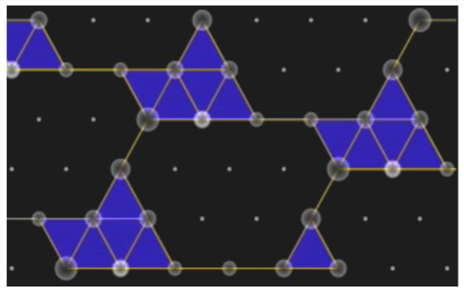
Methods

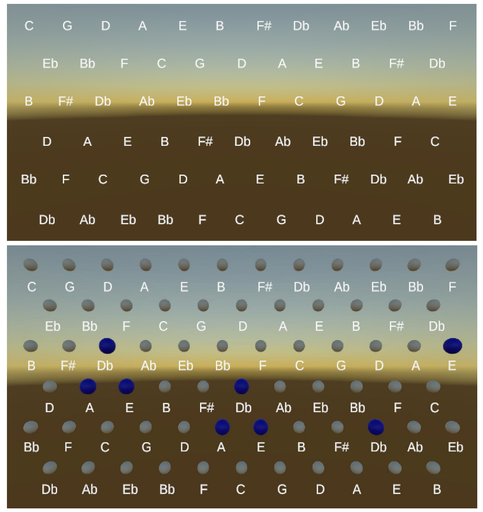
- **Unity [2]**
  - Game engine used to construct the isochord visualization
- **Maestro [3]**
  - Toolkit from Unity Asset Store, includes demos and MIDI files used in the project
  - Enables live playback of MIDI files, scripting
- **Isochord Grid**
  - Visible grid elements are sphere GameObjects rendered in Unity
  - Consists of a mesh [4] that constructs the vertices of the grid
    - allows for consistent placement of GameObjects on top of grid vertices
- **Note Identification**
  - Script accesses data for individual MIDI events, determines note based on numerical MIDI value & octave
  - Sphere GameObjects are rendered and placed on the grid according to note values, destroyed after full duration of MIDI events
- **Chord Analysis [5]**
  - Root of chord is found, compared to database of chord types in Cmaj to determine quality of chord
  - Currently can only detect key & first chord

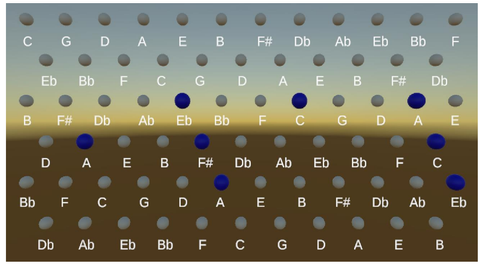
Results

- **Isochord Grid Visualization**
  - Left column pictures are from Bergstrom paper [1], right column pictures are Unity renders









  - In the Unity visualizations, triangles are major triads and diagonal lines are tritones
  - Harmonic relationships clearly visible

- **Grid Construction and Note Identification**
  - Grid is constructed at the start of the project
  - MIDI notes processed and identified at the start of the project, subsequently displayed on the grid at the appropriate times in the piece
    - MIDI is not being processed live, the data is simply taken from the input file, processed, and stored at the start of the project

Discussion and Future Work

- Work neglected:
  - Only individual notes are visible in the Unity render
    - Harmonic relationships can be inferred, but chords are not displayed more prominently (as seen in image from Bergstrom paper)
      - Triads are not shaded, nor are more complex chord shapes
      - Thirds and fifths are not connected with lines
- Future work:
  - Fix chord analysis system, implement in visualization
  - Visualize chords using a combination of meshes and other GameObjects

References

[1] T. Bergstrom et al. Isochords: Visualizing Structure in Music. *University of Illinois at Urbana Champaign*. DOI: 10.1145/1268517.1268565.

[2] Unity Engine. *Unity*. https://unity.com/.

[3] MusIT. Maestro – Midi Player Tool Kit – Free. *Unity Asset Store*. https://assetstore.unity.com/packages/tools/audio/maestro-midi-player-tool-kit-free-107994/

[4] J. Flick. Procedural Grid, a Unity C# Tutorial. *Catlike Coding*. https://catlikecoding.com/unity/tutorials/procedural-grid/.

[5] R. Jones. Midi Chord Analyzer, https://github.com/radionerd/Midi-Chord-Analyser/.

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